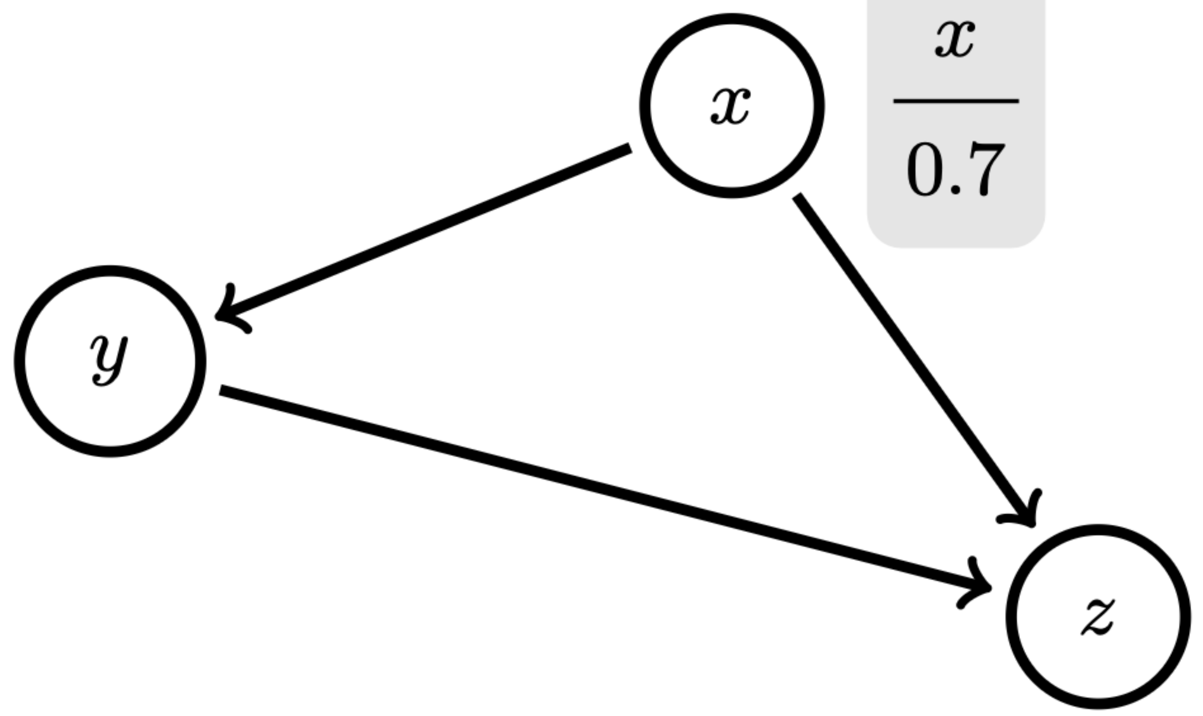
Probabilistic Ontologies

Bayesian Description Logic: $\mathcal{BEL}$

Probabilistic GCI: $\langle C \sqsubseteq D : x \rangle$

Crisp (classical) GCI: $\langle C \sqsubseteq D \rangle$

Bayesian Network



x
0.7

	y
x	1
$\neg x$	0.5

		z
x	y	0.3
x	$\neg y$	0.1
$\neg x$	y	0
$\neg x$	$\neg y$	0.9

Contexts determine necessary conditions:

$\langle$LuxuryHotel $\sqsubseteq$ $\exists$hasFeature.MeetingRoom : city$\rangle$
$\langle$LuxuryHotel $\sqsubseteq$ $\exists$hasFeature.SkiRentalService : ski_resort$\rangle$

OWL2 DL: $\mathcal{SROIQ}(\mathcal{D})$

Feature	DL Syntax	Example / Description
Atomic Concept	A	A named class (e.g., Person).
Top / Bottom	$\top$, $\bot$	The set of all individuals / the empty set.
Intersection	$C \sqcap D$	Things that are both C and D.
Union	$C \sqcup D$	Things that are either C or D (or both).
Negation	$\neg C$	Things that are not C.
Existential Restriction	$\exists R.C$	Has at least one R-relationship to C.
Universal Restriction	$\forall R.C$	All R-relationships must be to C.
Nominals ($\mathcal{O}$)	$\{ a_1, .. a_n \}$	A class defined by specific individuals.
Qualified Cardinality ($\mathcal{Q}$)	$\geq n\ R.C$, $\leq n\ R.C$	At least/most n relationships to class C.
Inverse Roles ($\mathcal{I}$)	R-	The reverse of a property (e.g., hasParent vs hasChild).
Role Inclusion ($\mathcal{R}$)	$R \sqsubseteq S$	If xRy, then xSy.
Complex Role Chains ($\mathcal{R}$)	$R_1 \circ \ldots \circ R_n \sqsubseteq S$	e.g., hasParent ∘ hasBrother ⊑ hasUncle.
Self-Restriction ($\mathcal{S}$)	$\exists R.Self$	Individuals related to themselves via R (e.g., Auto-regulating).
Transitivity ($\mathcal{S}$)	$Trans(R)$	If aRb and bRc, then aRc.
Symmetry / Asymmetry	$Sym(R)$ / $Asym(R)$	Defines bidirectional or strictly one-way links.
Reflexivity / Irreflexivity	$Ref(R)$ / $Irref(R)$	Whether aRa is always true or always false.
Disjoint Roles	$Dis(R, S)$	R and S cannot share the same pair of individuals.
Datatypes ($\mathcal{D}$)	D, $\exists T.d$, …	Integration of concrete data values (strings, integers).

OWL2 EL: $\mathcal{EL}^{++}$

Feature	DL Syntax	Example / Description
Atomic Concept	A	A named class (e.g., Person).
Top / Bottom	$\top$, ~~$\bot$~~	The set of all individuals / ~~the empty set.~~
Intersection	$C \sqcap D$	Things that are both C and D.
Union	$C \sqcup D$	Things that are either C or D (or both).
Negation	$\neg C$	Things that are not C.
Existential Restriction	$\exists R.C$	Has at least one R-relationship to C.
Universal Restriction	$\forall R.C$	All R-relationships must be to C.
Nominals ($\mathcal{O}$)	$\{ a \}$	A class defined by a single individual.
Qualified Cardinality ($\mathcal{Q}$)	$\geq n\ R.C$, $\leq n\ R.C$	At least/most n relationships to class C.
Inverse Roles ($\mathcal{I}$)	R-	The reverse of a property (e.g., hasParent vs hasChild).
Role Inclusion ($\mathcal{R}$)	$R \sqsubseteq S$	If xRy, then xSy.
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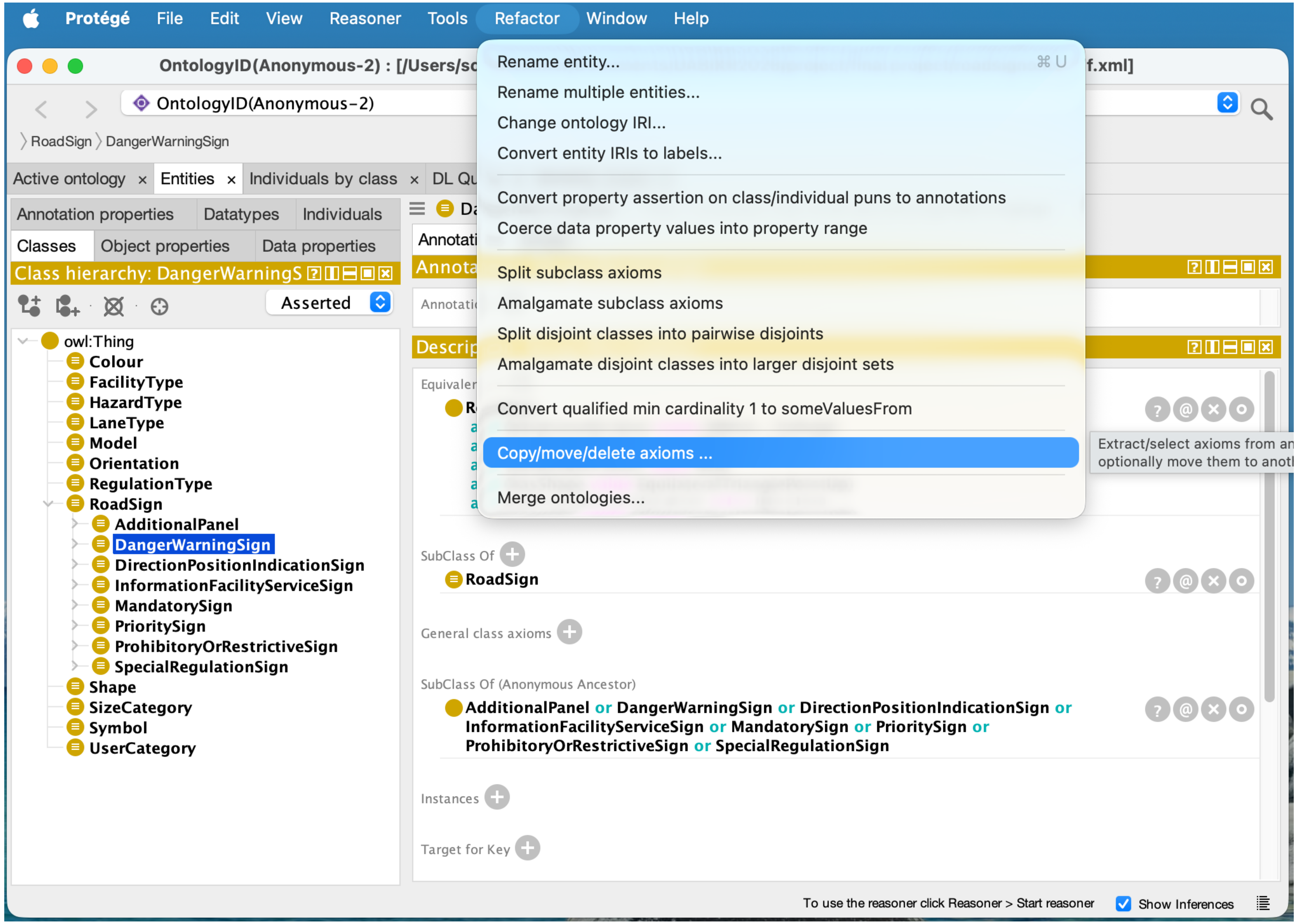
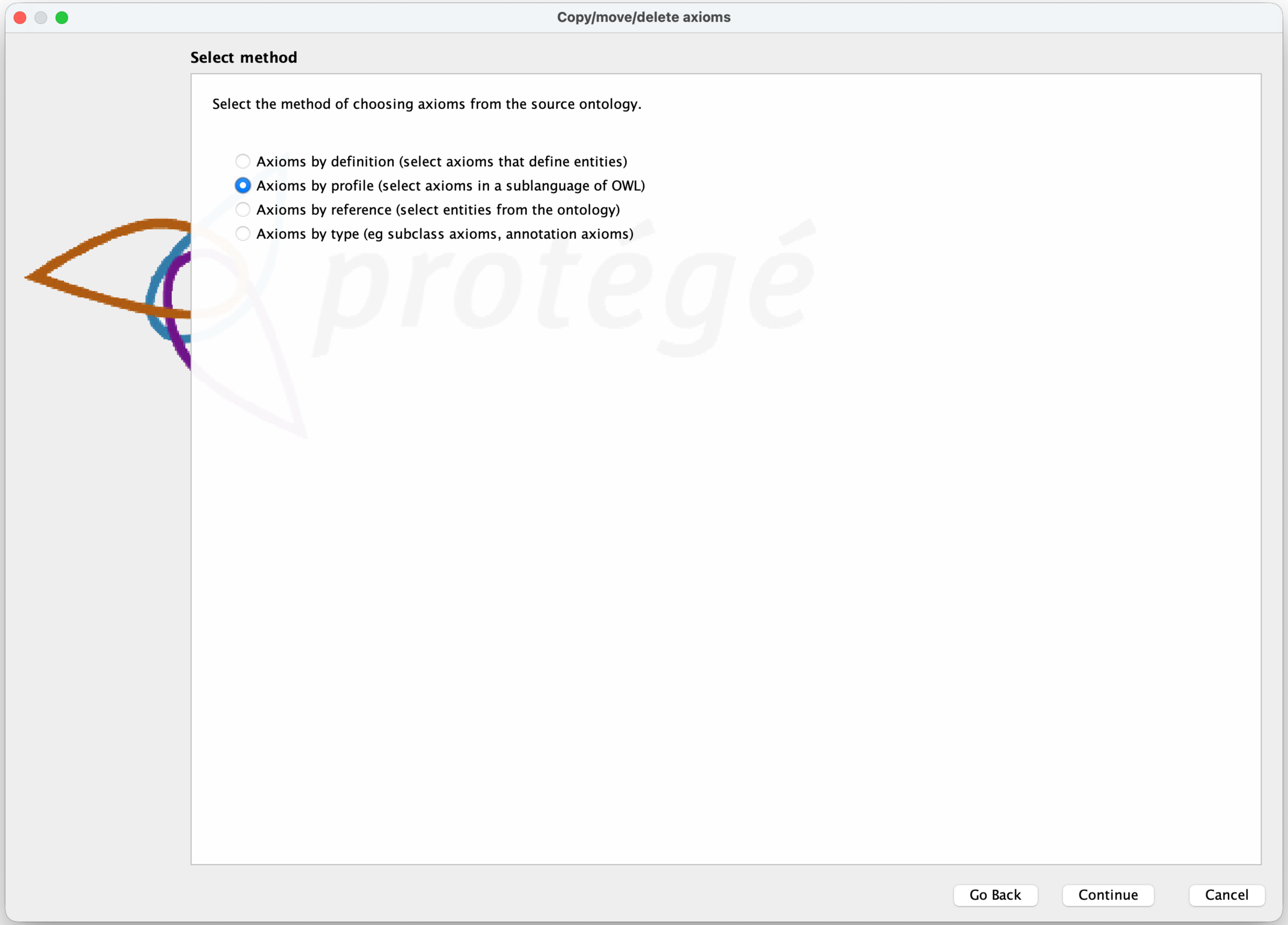
$\mathcal{EL}$

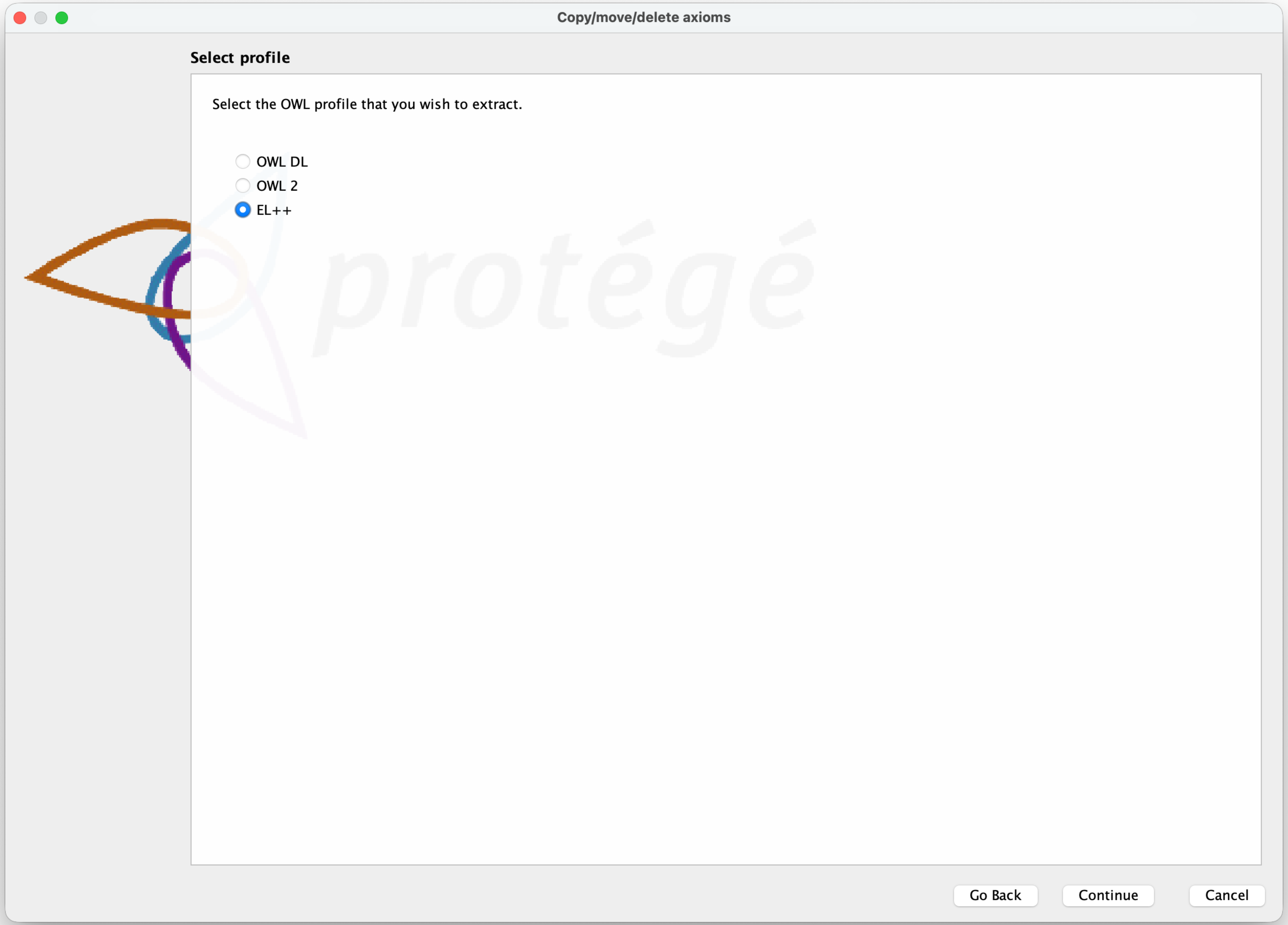
Extracting EL Profile



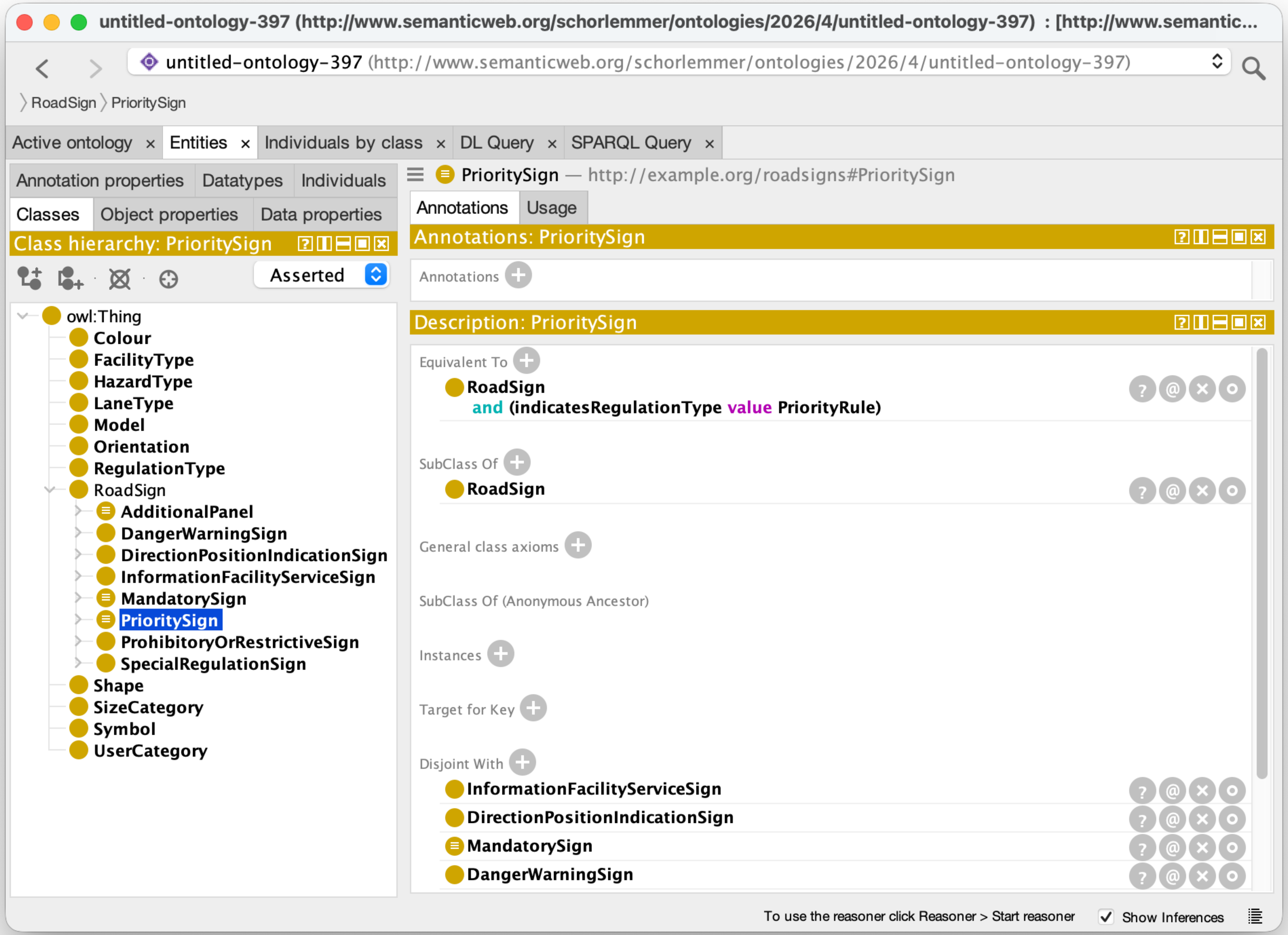
Extracting EL Profile



Extracting EL Profile



OWL2 EL Ontology ($\mathcal{EL}^{++}$)



Assignment

- Extract (or re-engineer) an $\mathcal{EL}$-fragment of your road-sign ontology
- You may focus only on a fragment of the road-sign taxonomy (e.g., subclasses of ProhibitoryOrRestrictiveSign)
- Adjust your necessary conditions (GCIs) to different contexts

Contexts for Road Signs

European Speed Limit Sign

US Speed Limit Signs